

39. DEVELOPMENT OF THE INTESTINAL LOOP

DURATION : 20:47

STUDY SHEET

COURSE SUMMARY

This video delves into the embryological development of the intestinal loop, focusing on the counter-clockwise rotation of the gut, the importance of the superior mesenteric artery, and its influence on intestinal growth dynamics. Key concepts covered include the relationship between the stomach, liver, and heart, as well as the various fulcrums essential for intestinal development and function. Additionally, the video explores the links between emotions and gut health, detailing how feelings like anger or fear can affect digestion. You will also learn the importance of considering peritoneal dynamics when treating intestinal disorders, thereby enriching your therapeutic approach in osteopathy.

DEVELOPMENT OF THE INTESTINAL LOOP

The **umbilical cord** is initially voluminous, but it will reduce in size as development progresses. Initially, the **intestine** develops outside the peritoneal cavity, due to a significant **trophic supply**, mainly provided by the **superior mesenteric artery**. This artery, which branches off from the aorta, forms the axis of intestinal rotation.

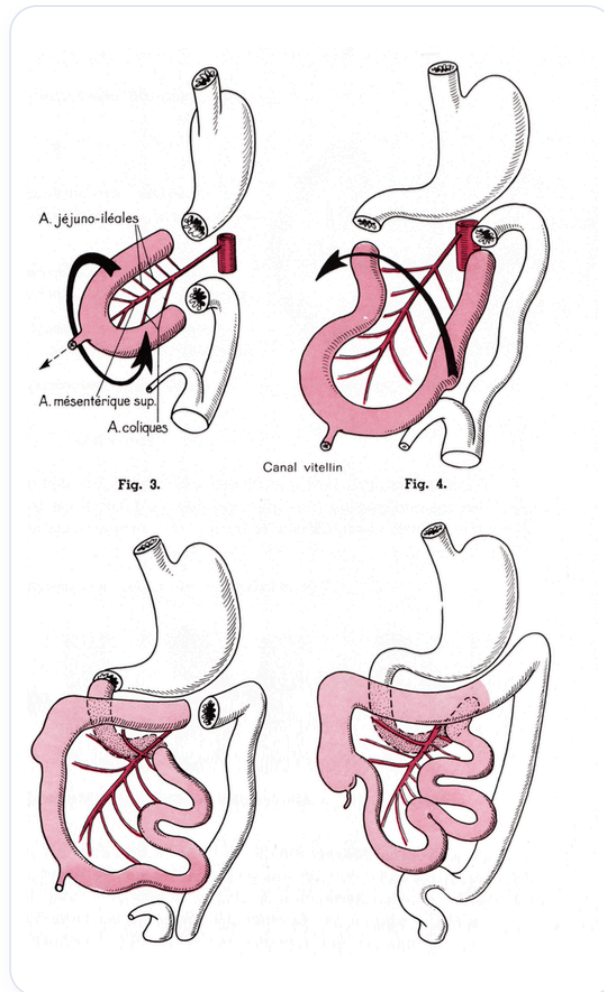


FIG. — *Intestinal loop rotation*

The rotation of the intestine occurs in a **counter-clockwise** direction, from front to back. In parallel, a **hepatogastric tube** and a digestive tract are established. A mesodermal structure is also observed, which is enveloped in tissue, corresponding to the **future task** at the umbilical level, the **omphalomesenteric canal**. Blood vessels develop further in this region.

There is a **synchronicity** between the stomach and the rotation of the intestine. When the stomach positions itself to the left, the liver moves to the right. This movement also leads to the heart being positioned to the left, causing splenic congestion. The **colon** begins to inflate, and the superior mesenteric artery becomes the fundamental fulcrum for intestinal development.

At the vertebral level, the superior mesenteric artery corresponds to the **T12-L1** axis, which is linked to the insertion zones of the diaphragmatic crura. This fulcrum is essential for the body and influences the growth dynamics of the intestine.

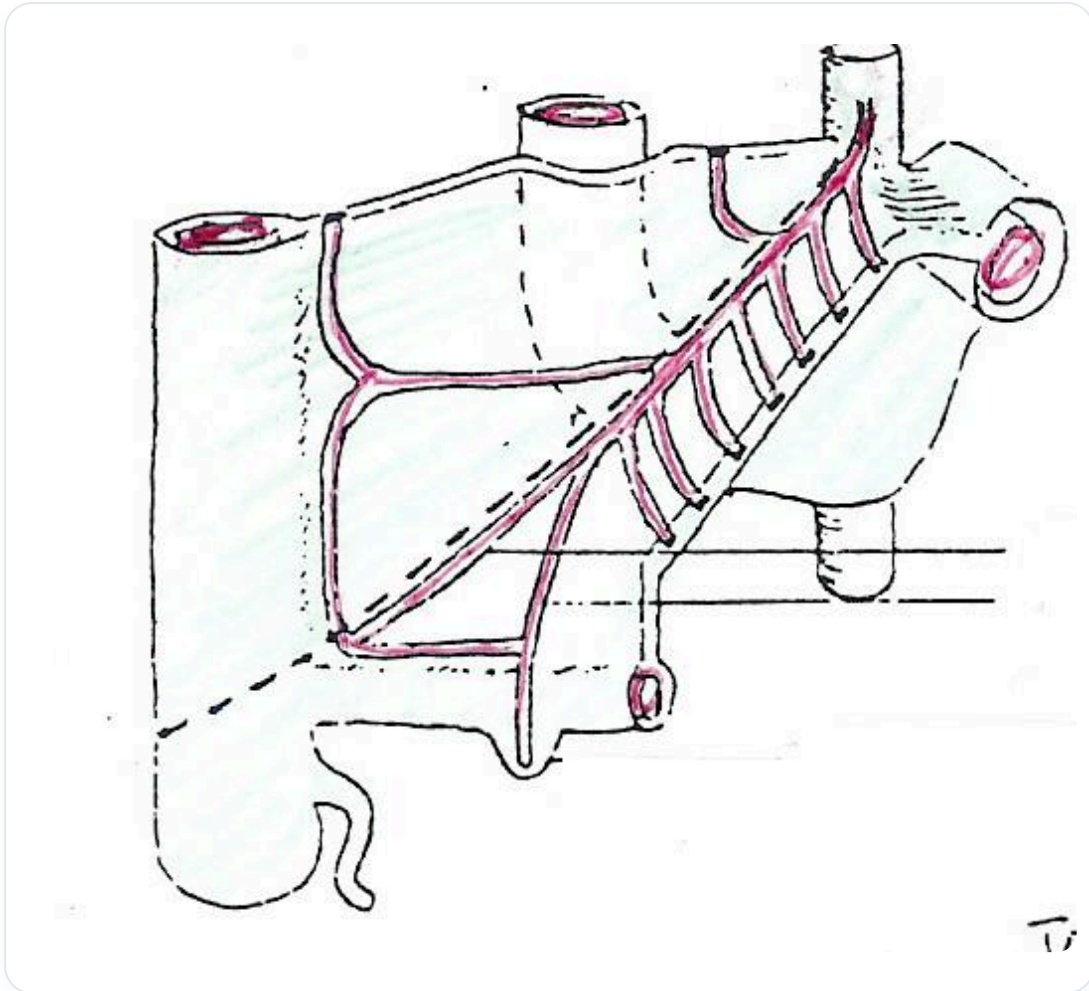


FIG. — Embryonic arterial trunks

The growth of the intestine is characterised by **differential speeds** linked to polarities. A greater metabolic concentration develops upwards, while the lower part grows less rapidly, thus forming the anlage of the **left colic flexure**. This movement of growth and rotation is crucial for the formation of the intestine.

By observing the dynamics of the intestine, several important fulcrums can be identified: the mesenteric artery, the left colic flexure, and the duodenal frame. These fulcrums are essential for the development and function of the intestine.

The **omphalomesenteric canal**, which is the axis of rotation, ends in a remnant called **Meckel's diverticulum**, located between the pubic axis and the caecum. Brain development is also dependent on the digestive vascular system. If the trophic supply is insufficient, this can lead to complications.

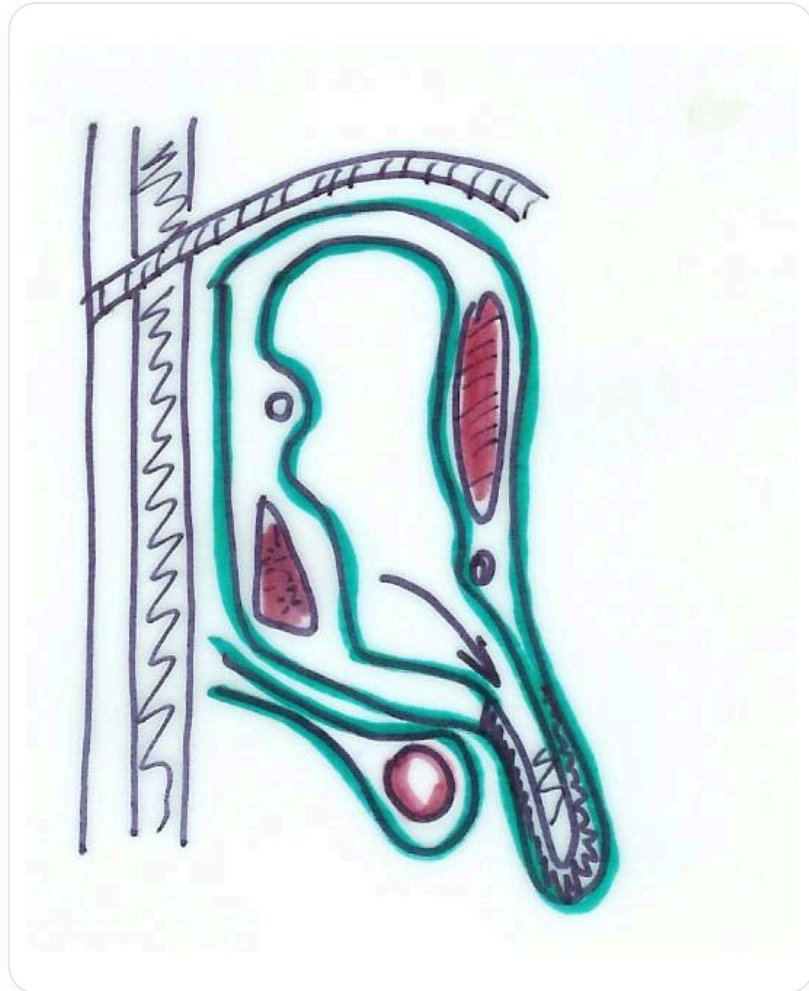


FIG. — Intestinal villi circulation

The re-integration of the intestine occurs during development, particularly around the third month of pregnancy, when the umbilical cord begins to form. This process is crucial for establishing connections with the mother, especially through the **amniotic cavity**, which allows for electrolyte and hormonal exchanges.

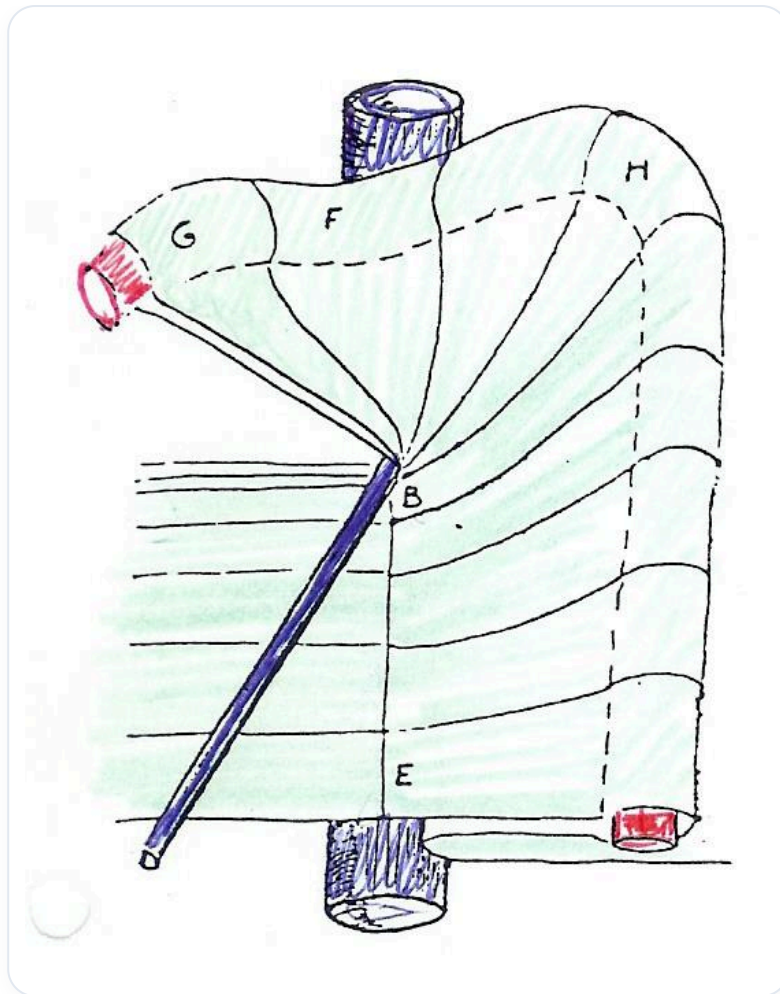


FIG. — Embryonic ventral/lateral fold

Emotions and states of mind are also linked to intestinal health. The **ascending colon** is associated with dietary injustices, while the **transverse colon** represents the moods of the soul. Emotions such as anger and fear can influence digestive health.

The **descending colon** is linked to family issues and the search for independence. The **appendix** and **caecum** play an important role in immune defence. Intestinal problems in children, such as **appendicitis**, can be linked to embryonic developmental phases.

The treatment of intestinal disorders must take into account peritoneal dynamics and the relationships between different organs. Stabilising the peritoneal frame can help resolve many health problems, including infections and inflammations.

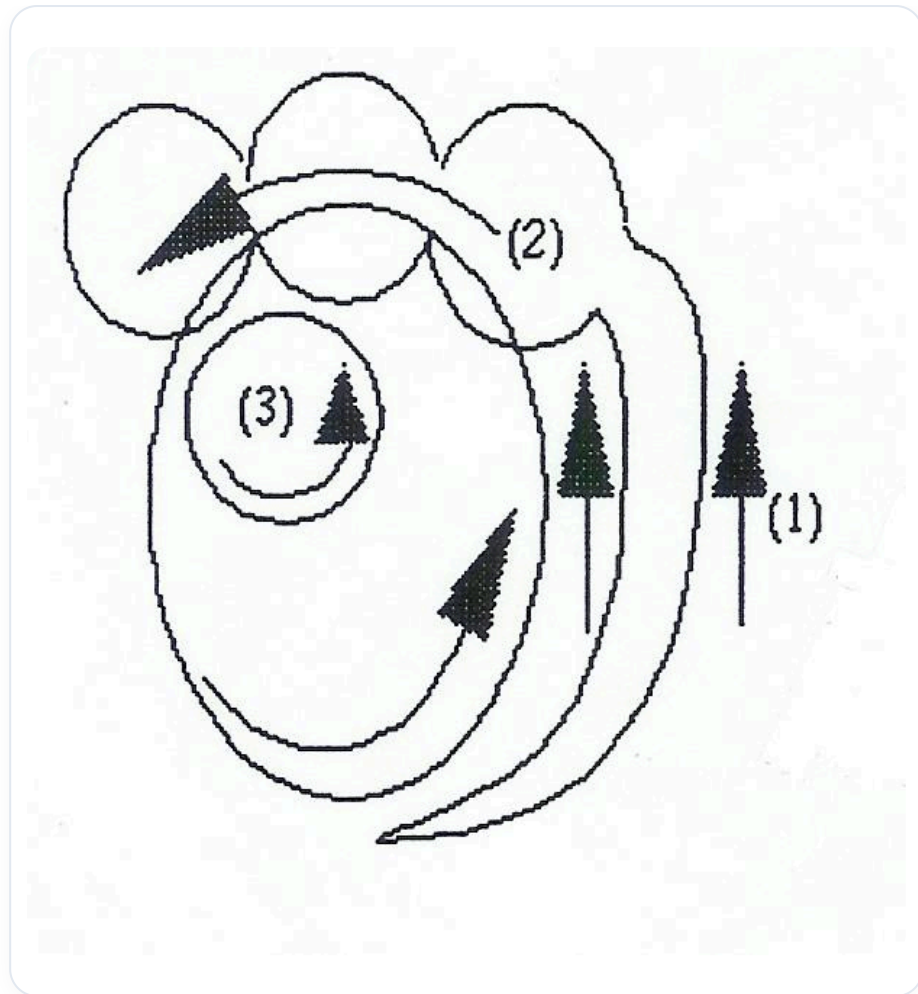


FIG. — *Embryological vortex*

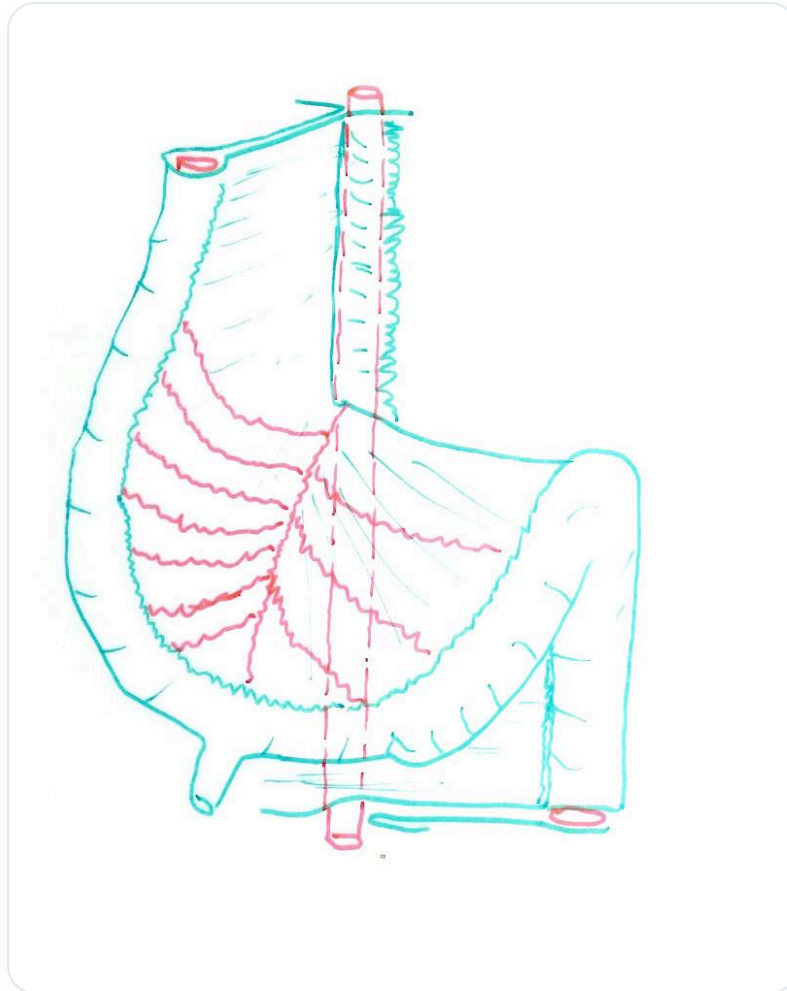


FIG. — *Embryonic digestive system*

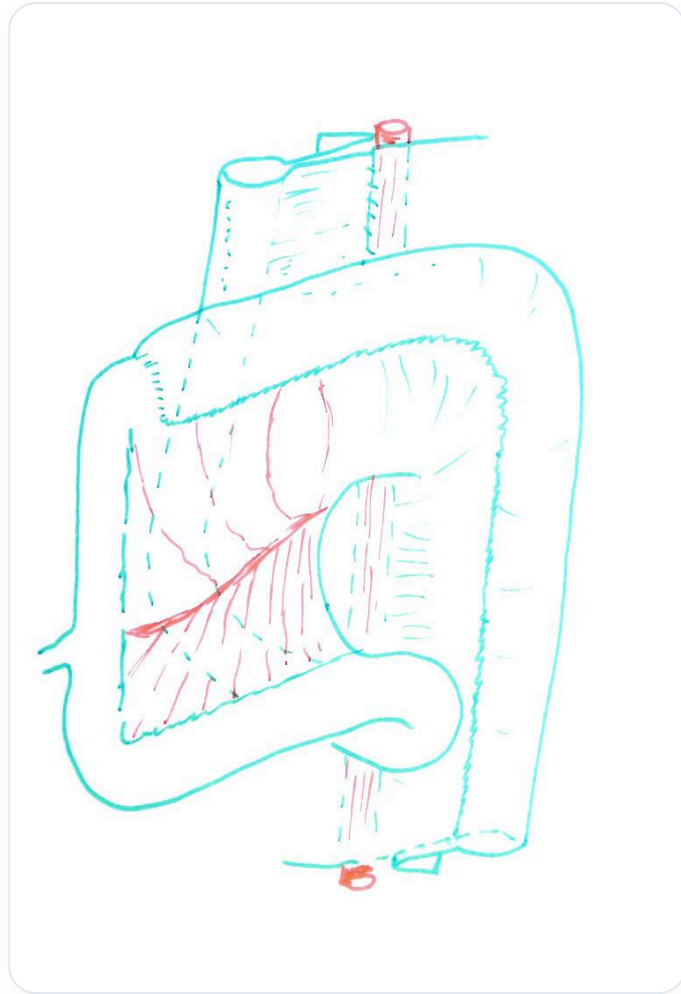


FIG. — Midgut, rotation

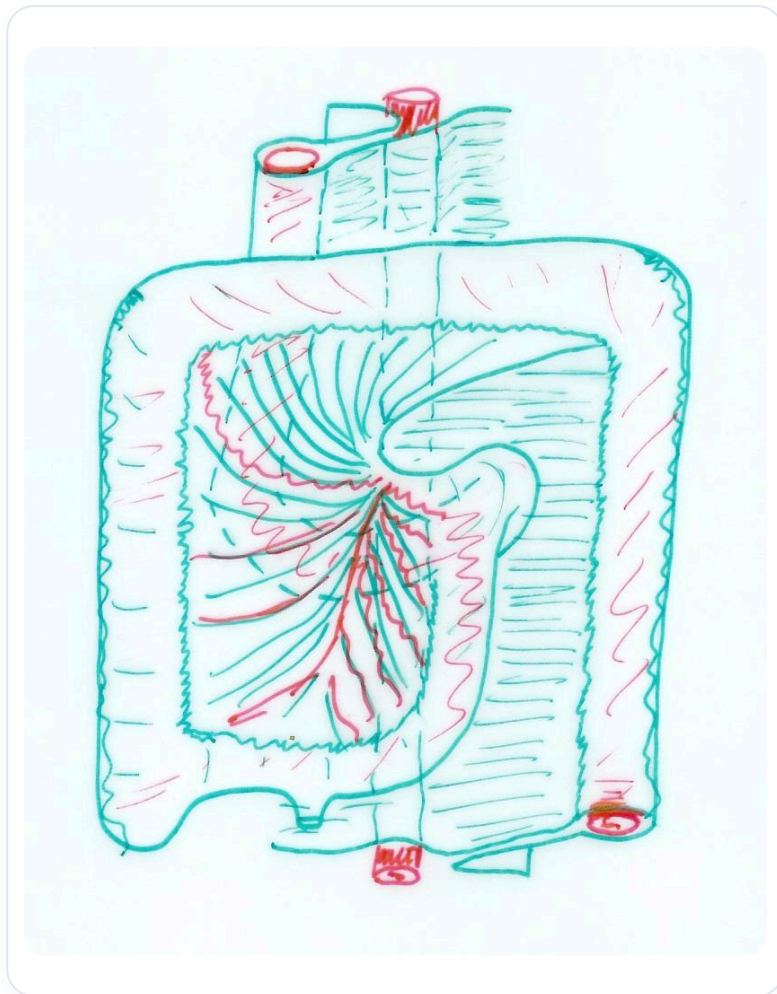


FIG. — Vascularized small intestine

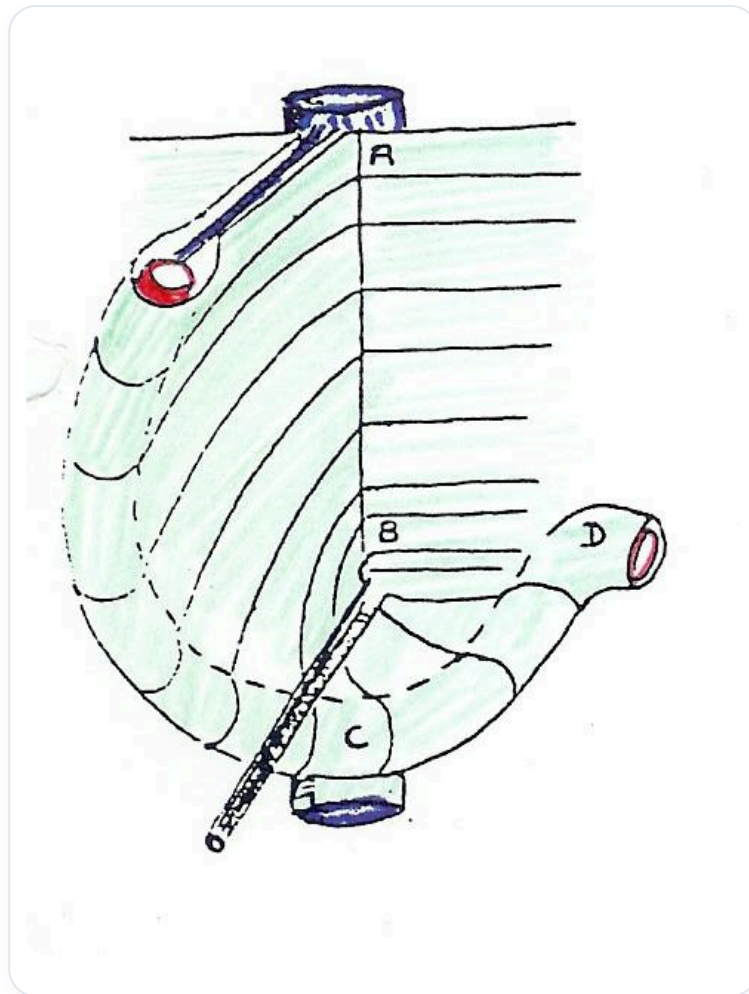


FIG. — Amniocentesis, Embryo, Uterus

Appendice vermiforme (A-C2). L'appendice prend naissance de l'*extrémité postéro-médiale du caecum*. Sa situation est très variable (**D**) : dans environ 65 p. 100 des cas, l'appendice est refoulé vers le haut derrière le caecum (*situation rétrocaecale haute*) ; dans 31 p. 100 des cas, elle dépasse la linea terminalis dans le petit bassin (*situation basse*) ; dans plus de 2 p. 100 des cas, il est horizontal derrière le caecum (*situation rétrocaecale transversale*) ; chez 1 p. 100 des hommes, il est refoulé vers le haut, en avant de l'iléon (*situation paracaecale, pré-iléale montante*) ; chez environ 0,5 p. 100, il est ascendant derrière l'iléon (*situation paracaecale, rétro-iléale montante*). Dans la **position la plus fréquente (rétrocaecale haute)**, la base de l'appendice vermiforme se projette sur la paroi abdominale antérieure, au **point de McBurney (E)**. Il se situe à la jonction entre le tiers externe et le tiers moyen d'une ligne tendue entre l'épine iliaque antéro-supérieure et l'ombilic. L'appendice fait en moyenne 10 cm de long et 6 mm d'épaisseur. Les trois ténias du caecum (**C**) partent en étoile de la base de l'appendice, dans la paroi duquel ils forment une *couche musculieuse longitudinale fermée*.

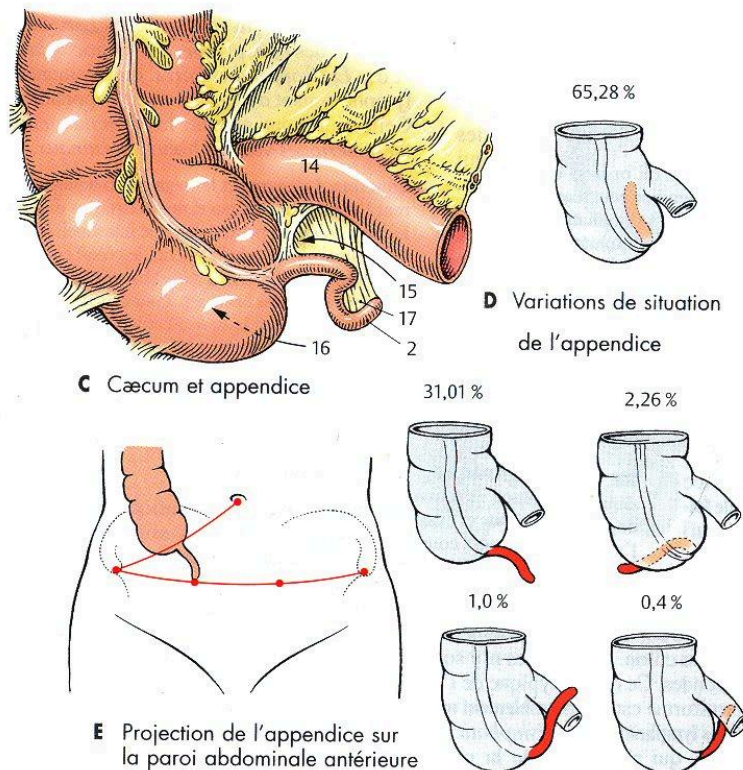


FIG. — Appendix, cecum, variations

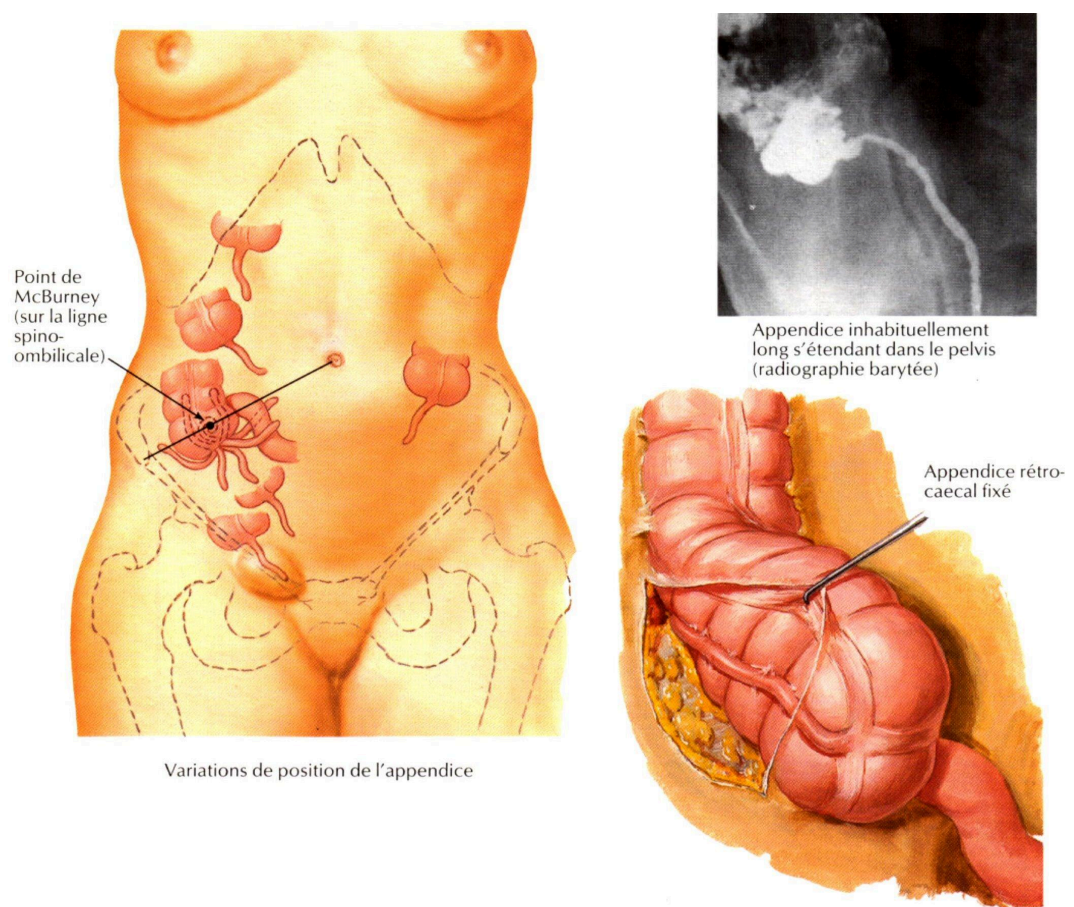


FIG. — Appendix position

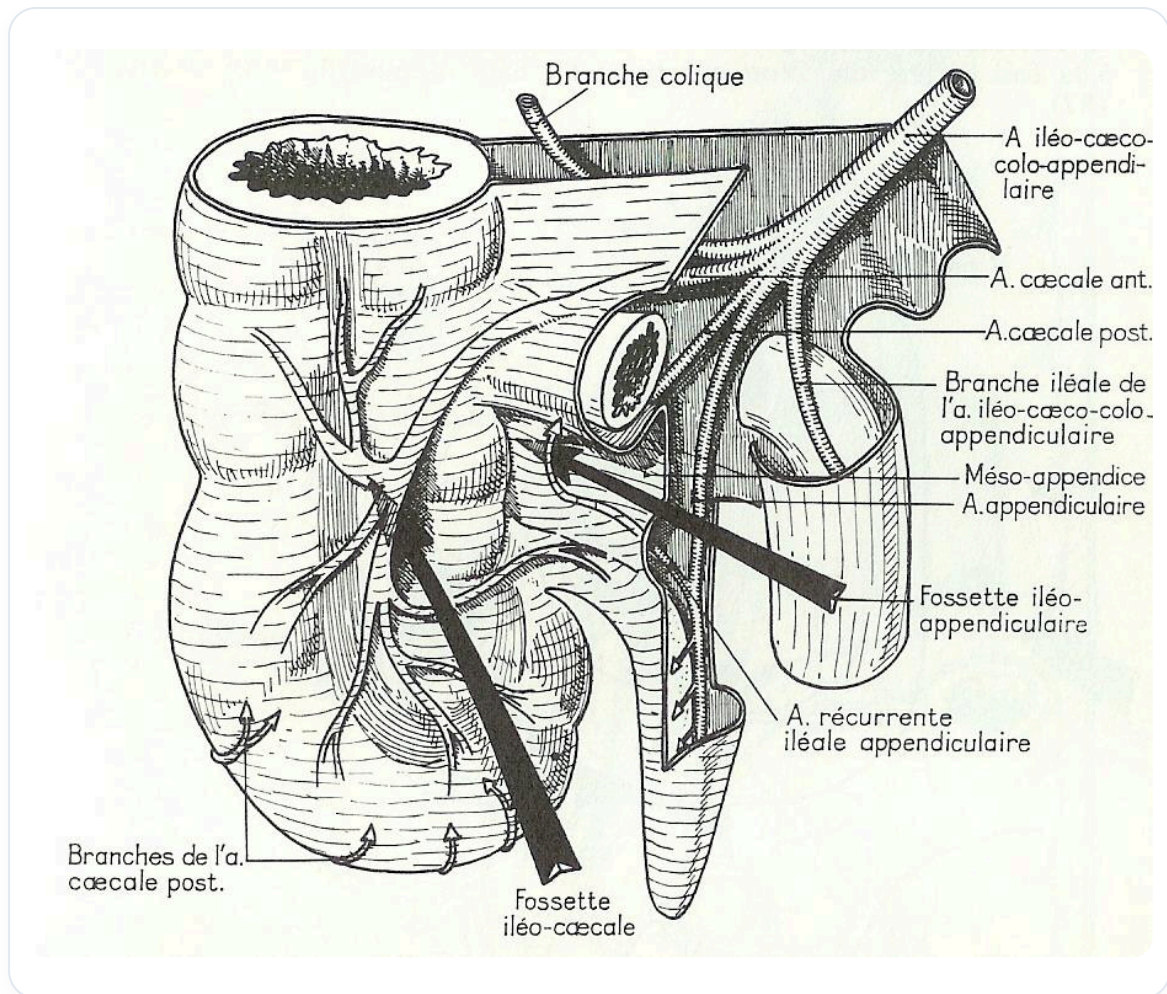


FIG. — Appendix cecum vascularization